

CHAPTER 3

Atmosphere and Climate

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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The Big Questions

1 What is the composition of the atmosphere?

✔ ANSWER

The atmosphere is a mixture of gases:

- **Nitrogen 78%** and **oxygen 21%** (the two main gases).
- Argon 0.93%, carbon dioxide 0.04%, others 0.03% (helium, neon, krypton, xenon, ozone, hydrogen).
- It also has water vapour (0.1 to 0.4 per cent) and tiny dust particles.

2 How do the different layers of the atmosphere affect the planet Earth?

✔ ANSWER

- **Troposphere:** air we breathe, clouds and nearly all weather (rain, fog, hail).
- **Stratosphere:** has the ozone layer that filters the Sun's harmful ultraviolet radiation.
- **Mesosphere:** most meteorites burn up here.
- **Thermosphere:** reflects radio waves back to Earth (helps radio transmission); auroras occur here.
- **Exosphere:** very thin air; light gases escape into space.

Together they protect life and shape weather and climate.

3 What is the mechanism of monsoon?

✔ ANSWER

Monsoon is the seasonal reversal of wind direction, caused by unequal heating of land and sea.

- **Summer (south-west):** land heats faster → low pressure over land, high pressure over the ocean → moist winds blow from sea to land and bring rain (June–September).
- **Winter (north-east):** land cools faster → high pressure over land → cold, dry winds blow from land to sea (October–February); they pick up moisture over the Bay of Bengal and give rain to Tamil Nadu and nearby areas.

4 How can we reduce our carbon footprint?

✔ ANSWER

- Walk, cycle or use public transport/carpool.
- Switch off lights and fans when not in use; use renewable energy.
- Use water carefully.
- Reuse, recycle and avoid single-use plastics.
- Plant and protect trees; live an eco-friendly lifestyle.

🤔 Think About It · Introduction

? Can you imagine what would happen if there were no atmosphere? Discuss your thoughts with your friends and teachers.

💡 SAMPLE ANSWER

- There would be no air to breathe, so no life.
- Harmful ultraviolet rays of the Sun would reach us directly.
- Days would be extremely hot and nights extremely cold, because no heat would be trapped.
- No clouds, rain or wind; no sound could travel; meteorites would hit the ground.

⭐ Let's Recall · Composition of the Atmosphere

? How is nitrogen useful for plants?

✔ ANSWER

Plants need nitrogen to make proteins and to grow. They cannot take it from the air directly; bacteria in the soil and in root nodules of pulses (e.g., Rhizobium) change it into a form plants can absorb.

🤔 Think About It · Elements of Weather and Climate

? Sometimes the wind is so strong that it is difficult to walk against it. Can you think of some other times when strong winds have caused problems for you?

💡 SAMPLE ANSWER

- Umbrella turned inside out in rain.
- Dust storm got into eyes; hard to ride a bicycle against the wind.
- Branches or trees fell on the road; electricity went off.
- Roof sheets, clothes on the line or shop boards blew away.

⭐ Let's Explore · Wind

? Find out other categories of wind based on speed and their common effects, and share your findings in class.

💡 SAMPLE ANSWER

Some other categories (Beaufort scale):

Wind	Speed (km/hr)	Common effects
Light air	1–5	Smoke drifts; wind vane does not move
Gentle breeze	12–19	Leaves and small twigs move; light flags open

Moderate breeze	20–28	Dust and loose paper rise; small branches move
Fresh breeze	29–38	Small trees sway
Near gale	50–61	Whole trees move; hard to walk against the wind
Gale	62–74	Twigs break off trees
Strong gale	75–88	Slight damage to buildings
Hurricane	118 and above	Very severe, widespread damage

⭐ Let's Analyse · Weather and Climate

? Note down the weather report from a local newspaper for two weeks and observe the changes occurring in the weather.

💡 SAMPLE ANSWER

Make a table with columns: Date | Max temp | Min temp | Humidity | Rainfall | Sky (sunny/cloudy).

Fill it daily for 14 days. Then note, for example: temperatures changed by a few degrees from day to day; cloudy days had higher humidity; rainy days were cooler.

This shows weather changes day to day, while climate is the average over many years.

🤔 Think About It · Seasons in India

? Hindustani Classical music associates certain rāgas with specific seasons. Find out which rāgas are connected to each season.

💡 SAMPLE ANSWER

Examples:

- **Rainy season (Varṣhā):** Megh, Miyan ki Malhar and other Malhar rāgas.
- **Spring (Vasanta):** Basant, Bahar, Hindol.
- **Summer (Grīṣhma):** Deepak.

Ask your elders and teachers for more.

⭐ Let's Explore · Monsoon

? Describe in your own words how monsoon affects the lives of the people around you.

💡 SAMPLE ANSWER

- Farmers sow crops like paddy when the monsoon rain comes; a good monsoon means a good harvest.
- Wells, ponds and rivers fill up, so there is more drinking water.
- Heavy rain causes waterlogging, traffic jams and sometimes floods; schools may close.
- It brings relief from heat; festivals like Teej are celebrated.

- Diseases like dengue and malaria spread more in this season.

⭐ Climate Change

? What actions can be taken to reduce such changes?

✔ ANSWER

- Reduce carbon footprint: save electricity, walk, cycle, use public transport.
- Use renewable energy (solar, wind).
- Protect forests and plant trees.
- Adopt sustainable lifestyles: reuse, recycle, avoid plastic.

⭐ Let's Explore · My Carbon Footprint

Step 1–2 **Tick the options that best describe your daily habits (Transport, Electricity use, Water use, Waste and plastics) and add up your score (Low = 1, Medium = 2, High = 3, Very high = 4 points).**

💡 SAMPLE ANSWER

Example:

- Transport: I use public transport → 2
- Electricity: I sometimes forget to switch off fans → 2
- Water: I use one bucket for bathing → 1

- Waste: I sometimes use plastic bottles → 2

Total = 7 points. A lower score means a smaller carbon footprint.

Step 3 **My Climate Action Pledge: write two simple changes you can make to reduce your score.**

💡 SAMPLE ANSWER

1. I will switch off lights and fans every time I leave a room.

2. I will carry my own cloth bag and steel water bottle instead of plastic.

💬 Classroom Discussion · Punjab Floods 2025: A Case Study

1 **To what extent did natural factors cause these floods compared to human activities?**

💡 SAMPLE ANSWER

- **Natural:** very heavy monsoon rain, made stronger by western disturbances; rain in Himachal Pradesh and Jammu & Kashmir; rivers Satluj, Beas, Ravi and Ghaggar already flowing high.
- **Human-made:** weak, old embankments (dhūsī bāndh); houses and farms too close to rivers; silt in rivers and dams; late or unclear warnings.

Nature started the flood, but human factors made the damage much worse.

2 **Do you think better planning could have reduced the damage? How?**

💡 SAMPLE ANSWER

Yes.

- Strengthen and repair embankments in time.
- Do not build houses and farms on river floodplains.
- Remove silt from rivers and dams regularly.
- Give early, clear flood warnings and keep relief camps ready.

3 **What are the guidelines for the management of floods according to the National Disaster Management Authority (NDMA)?**

💡 SAMPLE ANSWER

Some main do's and don'ts:

- **Before:** listen to warnings on radio/TV/mobile; keep an emergency kit (torch, food, water, medicines, documents); know the way to higher ground or relief shelter.
- **During:** move to higher places; switch off electricity and gas; do not walk or drive through flowing water; stay away from electric poles and wires.

- **After:** drink boiled or chlorinated water; avoid floodwater; return home only when officials say it is safe.

Planning steps: flood forecasting, embankments, flood-plain zoning, drainage.

4 **What role can students/youth play in helping with disaster preparedness?**

💡 SAMPLE ANSWER

- Spread awareness about warnings and safety rules.
- Learn first aid and take part in school mock drills.
- Help prepare family emergency kits.
- Volunteer in relief camps (collect food, clothes, medicines) under adult guidance.
- Help keep drains clean and plant trees.

❓ Questions and activities

1 **What is atmosphere? Explain its composition with the help of a pie diagram.**

✔ ANSWER

The **atmosphere** is the blanket of air surrounding the Earth, held by gravity. It is a mixture of gases vital for life.

Draw a circle and divide it as below (angle = % × 3.6°):

Gas	Share	Angle in pie
Nitrogen	78%	about 281°
Oxygen	21%	about 76°
Argon	0.93%	about 3°
Carbon dioxide	0.04%	very thin slice
Others	0.03%	very thin slice

It also has water vapour (0.1–0.4%) and dust particles.

2 **Draw a labelled diagram of the structure of atmosphere.**

💡 SAMPLE ANSWER

Draw layers from the ground upwards (see Fig. 3.3) and label:

- **Troposphere:** 0 to about 12 km (clouds, weather) → Tropopause
- **Stratosphere:** up to 50 km (ozone layer, aeroplanes) → Stratopause
- **Mesosphere:** up to 80 km (meteors burn) → Mesopause
- **Thermosphere:** 80 to 700 km (aurora, satellites, ionosphere)
- **Exosphere:** uppermost layer, very thin air

3 **Which are the four main seasons of India?**

✔ ANSWER

According to the IMD:

- **Winter:** December to early April
- **Summer or pre-monsoon:** April to June
- **Monsoon or rainy (advancing monsoon):** June to September
- **Post-monsoon (retreating monsoon):** October to December

4 **Why do you not feel the pressure of the atmosphere?**

✔ ANSWER

Air presses on us from all sides, and our bodies exert an equal counter-pressure from inside. The pressure inside our body is equal to atmospheric pressure, so the two cancel out and we do not feel it.

5 **In which layer of the atmosphere do aeroplanes fly and why?**

✔ ANSWER

Aeroplanes fly in the **stratosphere** because it is free of clouds and other weather disturbances, so flying is smooth and safe.

6 (a) **Distinguish between the troposphere and stratosphere.**

✔ ANSWER

Troposphere	Stratosphere

Lowest layer; average height about 12 km	Above the troposphere; up to 50 km
Temperature decreases with height	Temperature does not decrease with height
Has the air we breathe, water vapour and clouds	Free of clouds and weather disturbances
Nearly all weather (rain, fog, hail) occurs here	Has the ozone layer that filters UV rays; good for aeroplanes
Ends at the tropopause	Ends at the stratopause

6 (b) **Distinguish between the south-west monsoon and north-east monsoon.**

✔ ANSWER

South-west monsoon	North-east monsoon
Summer monsoon, June to September	Winter monsoon, October to February
Land is hotter → low pressure over land	Land is cooler → high pressure over land
Moist winds blow from sea to land	Cold, dry winds blow from land to sea
Brings most of India's rainfall	Little rain in most parts; rain on east coast (Tamil Nadu, Andhra Pradesh, parts of Karnataka) after crossing the Bay of Bengal

7 **Do it yourself: Convert the data of Table 3.3 into 'temperature and rainfall' graphs for the 10 stations (like Fig. 3.14 for Delhi) and arrive at some broad generalisations.**

💡 **SAMPLE ANSWER**

For each station: months on the x-axis; draw rainfall as bars (cm, right axis) and temperature as a line (°C, left axis).

Broad generalisations:

- Most stations get most rain in June–September (south-west monsoon).
- Chennai gets most rain in October–December (north-east/retreating monsoon).
- Places near the sea and equator (Thiruvananthapuram, Mumbai, Chennai) have small temperature ranges.
- Inland northern places (Delhi, Jodhpur) are very hot in summer and cool in winter.
- High places are cooler (Shillong, Leh); Leh is below 0°C in winter.
- Rainfall varies from 8.5 cm (Leh) to 225.3 cm (Shillong).

7.1 **Re-arrange the 10 stations according to their distance from the equator.**

✔ **ANSWER**

Nearest to farthest:

1. Thiruvananthapuram (8°29'N)
2. Bengaluru (12°58'N)

3. Chennai (13°4'N)
4. Mumbai (19°N)
5. Nagpur (21°9'N)
6. Kolkata (22°34'N)
7. Shillong (24°34'N)
8. Jodhpur (26°18'N)
9. Delhi (29°N)
10. Leh (34°N)

7.2 (a) **Find out two stations with the most extreme climate.**

✔ ANSWER

Annual temperature range = hottest month − coldest month:

- Leh: 17.2 − (−8.5) = 25.7°C
- Jodhpur: 33.9 − 14.9 = 19.0°C
- Delhi: 33.3 − 14.4 = 18.9°C (almost the same as Jodhpur)

Leh and Jodhpur (Delhi is a close third).

7.2 (b) **Find out two stations influenced by retreating monsoons.**

✔ ANSWER

Chennai (Oct 30.6 cm, Nov 35.0 cm, Dec 13.9 cm) and **Thiruvananthapuram** (Oct 27.3 cm, Nov 20.6 cm) get heavy rain in October–November.

7.2 (c) **Find out the two hottest stations in the months of (i) February (ii) June.**

✔ ANSWER

(i) February: **Thiruvananthapuram (27.3°C)** and **Chennai (25.7°C)**.

(ii) June: **Jodhpur (33.9°C)** and **Delhi (33.3°C)**.

7.3 (a) **Why does Shillong experience more rainfall than Kolkata?**

✔ ANSWER

Shillong is on the hills of Meghalaya at 1461 m. Moist monsoon winds from the Bay of Bengal are forced to rise over these hills; they cool and give heavy rain (mountains affect precipitation). Kolkata is on flat land near sea level, so the winds pass over without rising as much.

Shillong 225.3 cm, Kolkata 162.5 cm.

7.3 (b) **Why does Delhi receive more rainfall than Jodhpur?**

✔ ANSWER

Monsoon winds from the Bay of Bengal move north-west and lose moisture as they travel. Delhi still gets some of it, but by the time they reach Jodhpur (in the Thar desert, further west) little moisture is left. The Arabian Sea winds also pass Rajasthan along the Aravallis without being forced to rise.

Delhi 67.0 cm, Jodhpur 36.6 cm.

7.4 (a) **Now think why Thiruvananthapuram has an equable climate.**

✔ ANSWER

It is closest to the equator (8°29'N) and lies on the coast. The sea keeps temperatures even all year: 26.2°C to 28.7°C, a range of only 2.5°C.

7.4 (b) **Why does Chennai have more rainfall only after the fury of the monsoon is over in most parts of the country?**

✔ ANSWER

Chennai gets little rain from the south-west monsoon. In October–December the north-east (winter/retreating) monsoon winds pick up moisture over the Bay of Bengal and bring rain to the Tamil Nadu coast. So Chennai's heaviest rain (Oct 30.6, Nov 35.0 cm) comes after the monsoon has ended elsewhere.

7.4 (c) Why does Leh have moderate precipitation almost throughout the year?

✔ ANSWER

Leh is a cold desert high in the Himalayas (3506 m). The mountains block the moist monsoon winds, so it gets no heavy monsoon rain. It gets small amounts (0.5–1.3 cm) almost every month, mostly as snow brought by western disturbances in winter and a little rain in summer. Total only 8.5 cm a year.

7.5 Despite these differences across regions, can you observe any substantial evidence to conclude that the monsoons provide a very strong framework, lending overall climatic unity to the whole country?

✔ ANSWER

Yes.

- Almost all stations get most of their yearly rain in the monsoon months (June–September); e.g., Mumbai 50.6 + 61.0 + 36.9 + 26.9 cm, Delhi 7.4 + 19.3 + 17.8 + 11.9 cm.
- Where it is not so, the retreating/north-east monsoon gives the rain (Chennai, October–December).
- Winter months (December–February) are dry almost everywhere.

- The same rhythm of seasons (hot summer, rainy monsoon, dry winter) is seen across India. So the monsoon gives the country a common climatic pattern.

8 **Collect pictures of houses and clothing of people from different regions of India. Examine whether they reflect any relationship with the climatic conditions or the relief of those regions.**

💡 **SAMPLE ANSWER**

Yes, they are related:

- **Rajasthan (hot, dry):** thick mud walls and small windows keep houses cool; loose white cotton clothes and turbans protect from sun.
- **Kerala and Assam (heavy rain):** sloping roofs; houses on stilts in Assam against floods; light cotton clothes.
- **Ladakh and Himachal (cold, hilly):** stone/wood houses with thick walls, flat roofs in dry Ladakh, sloping roofs where snow falls; woollen clothes like pheran and shawls.
- **Plains (hot summer, cool winter):** cotton in summer, woollens in winter.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**